

assignments in

Junior Science

book 3
Chemistry

SUADA BILALI
CHRIS GREEF



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CHRIS GREEF**



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Editor: Shelley Barons
Design and formatting by DiZign Pty Ltd
Diagrams by DiZign Pty Ltd
Printed in Australia by Ligare Book Printer

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ISBN 978 1 92209 078 2

Introduction



Each book in the *Assignments in Junior Science* series provides a selection of worksheets covering one or two major topics. The worksheets are independent, supplying most information necessary to formulate answers. In the answer section at the back of each book, sample answers to most tasks have been given.

Many worksheets have an emphasis on literacy, others concentrate on problem solving, while others utilise science in an experiment or a game format. They are presented in a student-friendly manner and could be used to supplement science lessons for regular class teachers or casual teachers.

Suada Bilali and Chris Greef are both experienced and practising science teachers who had a lot of fun putting together this collection of worksheets.

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History of the atomic model

1a



- 1 Read the following text about scientists and cross out the incorrect verb alternatives. The verbs are underlined. The first one has been done for you. Remember that historical recounts should be written in past tense.

People often ~~made up~~ / make up models to explain / to explained things. Scientists perform / performs experiments which show / shows them how something acts, or what something looks / looked like. This information is used to develop / to develops a model. The model of the atom has underwent / has undergone many changes.

The idea of atoms is suggested / was suggested by Democritus in ancient Greece (460 – 370 BC). He logically deducted / deduced that matter could not continue to be cut into smaller pieces forever, and that eventually there had to be / have to be a smallest particle, which he calls / called the atom (about 400 BC).

In England, John Dalton (1766 – 1844) heard / heard about the experiments of other chemists. From their results he concluded / concludes that the atoms of the same element was / were identical and cannot be / could not be broken. In a chemical reaction, the atoms were separated, joined or rearranged. Dalton's model of the atom were / was like a ball (1803).

When English scientist Joseph Thomson (1856 – 1940), and other scientists, experiment / experimented with electricity, they finded / found that cathode rays are / were made up of tiny particles which travelled / travel in straight lines and carried / carryed a negative electric charge. These cathode rays came / comed from the negative terminal (ie the cathode) and travelled to the positive terminal (ie the anode), when very high voltages have / were applied. Thomson concludes / concluded that the rays make up / were made up of electrons, which were very small (they was / were 2000 times lighter than a hydrogen atom) and had / have a negative charge. But if negative electrons came / come from neutral and relatively large atoms, the atom had to / has to look like a plum pudding: the dough of the pudding was a uniform substance with a positive charge and contained / contain tiny sultanas with a negative charge (1904).

- 2 In the following text about atoms and radioactivity, there are 12 spelling mistakes. Underline the incorrectly spelt words and write them correctly in the margin.

Radioactivity had been discovered and alpha particles were identified as tiny positive protons. A student of Joseph Thomson, the New Zealand born Ernest Rutherford (1871 – 1937), shot a beam of alpha particals at a very thin sheet of gold foil. He expected the protons to fly straight threw the foil, practically uneffected by the positive and negative particles in the gold atoms. Rutherford was suprised to find that some protons were deflected, some even bounced back to the sauce. Based on his measurements, Rutherford concluded that the atom's protons had to be concentrated in a small nucleus leaving the electrons to suround it. The mass of the atom was determined by the nucleus, which was much heavier than the electrons, and its size was determined by the electrons, which moved around the nucleus. Rutherford's model of an atom looked like a spherical cloud of negative electrons with a positive nucleus in the centre. If the atom was as big as a football feild, the nucleus would be as big as a tennis ball (1911).

The German Max Planck (1858 – 1947) formulated the quantum theory: Energy is given out or taken in by atoms only in lots (called quanta). Niels Bohr from Denmark (1885 – 1962) concluded that electrons jumped from one energy level to another, each jump being worth one qantum of energy. As a result the atomic model became like a soler system: a small nucleus, containing the heavy positive protons, was surrounded by several orbits on which the tiny negative electrons circled the nucleus (1913).

Erwin Schrodinger (1887 – 1961) and Wolfgang Pauli (1900 – 1958), both from Austria, proposed another model for the atom: The electron shells contained subshells of various shapes, each with a maximum of two electrons. But it was impossible to state where a particular electron was at any time; one could only say where an electron was most likley to be. Thus the atom appered like a nucleus inside many electron clouds (1926).

When the English scientist James Chadwick (1891 – 1974) discovered the existance of neutrons, which were as heavy as protons but carried no charge, the atomic model changed to include neutrons inside the nucleus together with the protons (1932).

History of the atomic model

1b



+

3 The following words are snippets from sentences which summarise the text from questions 1 and 2. Sort the snippets from each sentence and then write down the sensible sentences.

a | atom | Democritus from ancient Greece | named | in about 400BC | the smallest particle |.

b | believed | in 1803 | like a ball | that | the atom | The English scientist John Dalton | was |.

c | drew | In 1904 | Joseph Thomson from England | like a positively charged plum pudding with negative plums in it | the atom |.

d | and | concentrated | Ernest Rutherford | In 1911 | in the central nucleus | moved | the negative electrons | onto a cloud surrounding the nucleus | the positive charge |, who was born in New Zealand, |.

e | but | in 1913 | kept | onto several orbits | placed | rather like our solar system | The Danish Niels Bohr | the negative electrons | the positive nucleus |.

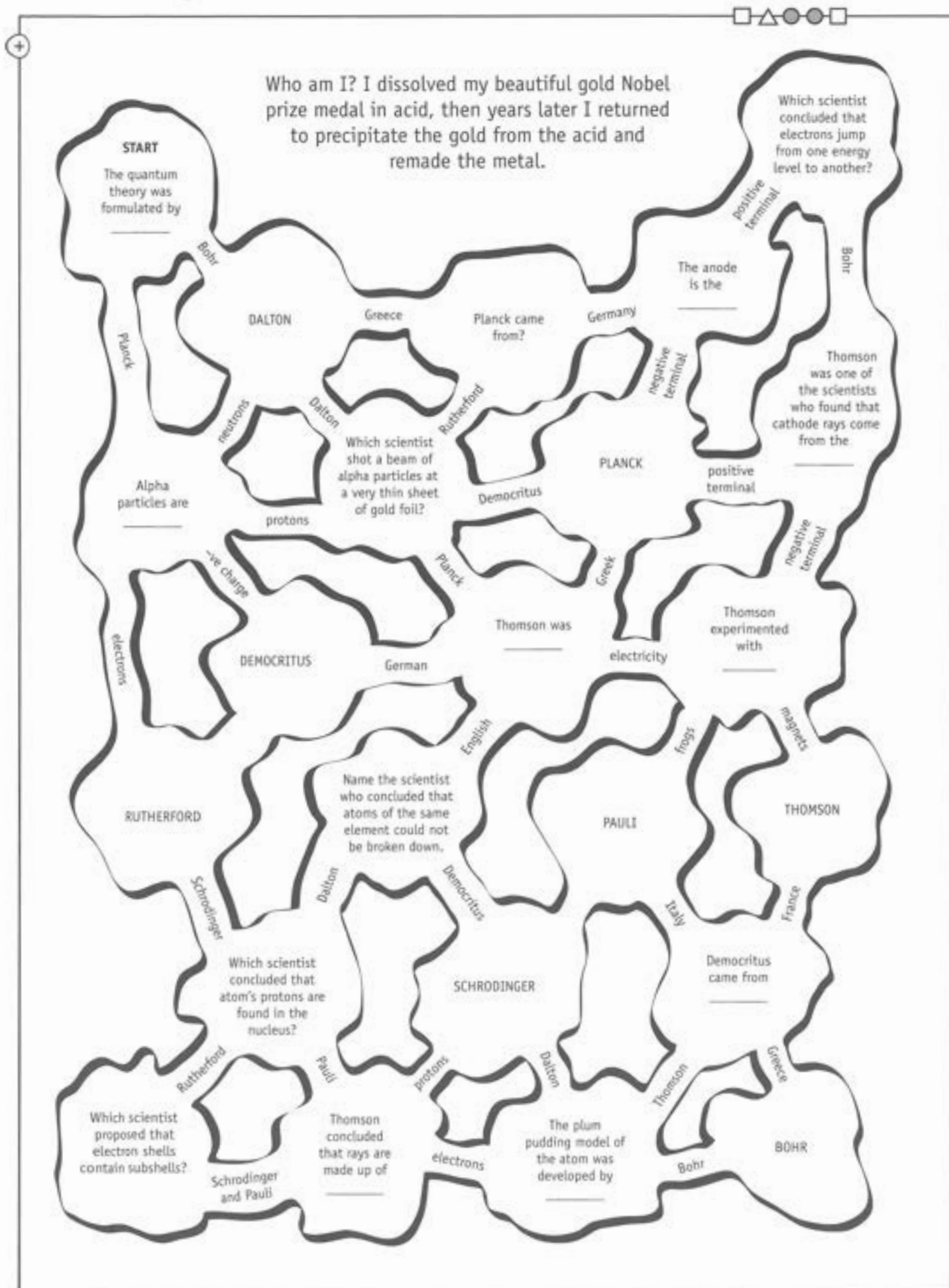
f | an atom where the nucleus was surrounded by several orbital clouds | and | combined | developed | in 1926 | The Austrian scientists Erwin Schrodinger and Wolfgang Pauli | the idea of orbits and clouds |.

g | discovered | In 1932 | inside the positive nucleus | neutral neutrons | the English scientist James Chadwick |.

Drawing atomic models

In the paragraphs above various scientists developed certain models of atoms. Each model explained what the scientists knew at that time. New discoveries made new atomic models necessary.

4 Draw each model as described in the text. Next to your diagram write the description of the model as well as the name of the scientist and the year of discovery.



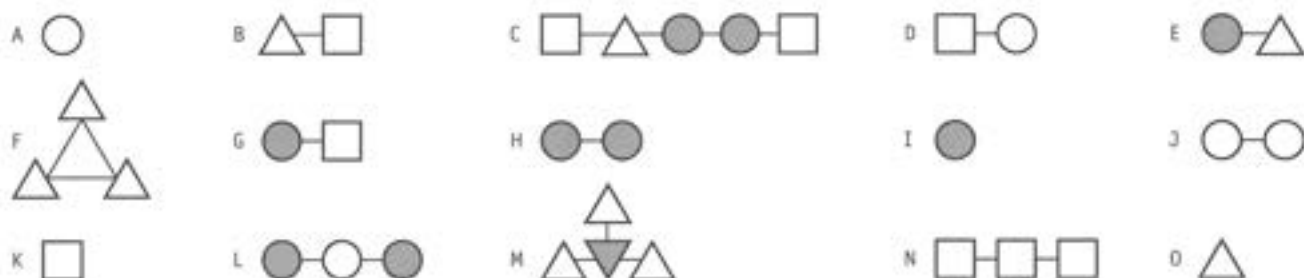


Matter can be divided into groups using certain criteria. Matter can be divided into living and non-living things. Matter can be divided into solids, liquids and gases. Matter can be grouped into metals and non-metals. Matter can be split into single particles and multiple particles. Matter can be divided into substances which contain one kind of atom or more than one kind of atom.

Definitions

- **Atoms** are tiny particles. They are the simplest form of matter. Atoms are made up of single particles.
- **Molecules** are also tiny particles. They are made up of two or more atoms. These atoms are held together by chemical bonds.
- **Elements** are substances which contain only one kind of atom.
- **Compounds** are substances which contain particles that are made up of two or more different atoms.

Look at the following diagrams showing various particles and substances. Then answer the questions below by referring to the definitions.



1 Identify all atoms and all molecules and record them in the table.

Particles	Atoms	Molecules
Diagrams		

2 Identify all elements and all compounds and record them in the table.

Substances	Elements	Compounds
Diagrams		

3 Complete the following table about particles and substances by adding as many examples of each as possible from the diagrams.

Atom of element	Atom of compound	Molecule of element	Molecule of compound

4 Explain why one cell in the table from question 3 is left empty. _____

5 Make up two more examples for each cell in question 3 and draw them below the table.

6 Explain what the lines in the diagrams represent. _____



Archimedes of Syracuse (287 – 212 BC) ran naked out into the streets crying 'Eureka!', excited after finding out the answer to King Hieron II's problem.

1 The following words relate to this topic about matter but are incomplete. Identify the missing letters and then use them and their matching numbers to complete the sentence below and discover the problem King Hieron had.

ma_te_	ato_s	s_bstan_e	mol__ule_	_a_tic_es
1 6	2	3 5	4 5 8	17 6 15
_o_l_d	b_n_s	_or_	_riteri_	m_ta_s
8 7	11 13	9 2	5 10	4 15
l_qu_d	s_lid	g_s	co_poun_s	_ater
7 7	11	10	2 13	12
sil_e_	_ydr__en	_le_e_t		
14 6	16 11 19	4 2 18		

King Hieron II's problem

13 7 13	1 16 4	19 11 15 13 8 2 7 1 16 8	3 8 4	10 15 15	1 16 4	19 11 15 13	11 6	6 4 17 15 10 5 4
k								
8 11 2 4	11 9	7 1	12 7 1 16	8 7 15 14 4 6	12 16 4 18	2 10	7 18 19	1 16 4
19 11 15 13 4 18								
5 6 11 12 13								
?								

2 Write each of the words listed above into a sentence about science.

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

3 Research: Use library resources to find out how Archimedes solved the king's problem. This should explain why he was naked at the time!

The periodic table of elements

5a



+

A brief history

In 1829 scientists knew about 40 elements. German scientist Johann Dobereiner noticed that lithium, sodium and potassium had very similar properties. Calcium, strontium and barium were also similar to each other. Chlorine, bromine and iodine formed another triad. In 1864, English scientist John Newlands replaced the triads with a law of octaves, which was based on the atomic weights of 60 known elements as well as their properties. Dmitri Mendeleev, from Russia, refined the periodic table in 1869, leaving gaps for some elements which had not been discovered. Mendeleev predicted the properties of some of those missing elements quite accurately. However, there were some inconsistencies: certain elements were too heavy for their position in the periodic table. Henry Mosely (England, 1914) resolved these problems, when he based the order of the elements on their atomic number rather than their weight. Today we know 103 elements and they are all organised into the periodic table of elements.

Some properties of the first 20 elements

Name	Symbol	Atomic weight	Order by weight	Atomic number	Atomic radius (pm)	Melting point (°C)
Aluminium	Al	27		13	143	660
Argon	Ar	40		18	94	-189
Beryllium	Be	9		4	112	1280
Boron	B	11		5	88	2030
Calcium	Ca	40		20	197	838
Carbon	C	12		6	77	3730
Chlorine	Cl	35		17	99	-101
Fluorine	F	19		9	64	-220
Helium	He	4		2	50	-270
Hydrogen	H	1		1	37	-259
Lithium	Li	7		3	152	1650
Magnesium	Mg	24		12	160	650
Neon	Ne	20		10	60	-249
Nitrogen	N	14		7	70	-210
Oxygen	O	16		8	66	-219
Phosphorus	P	31		15	110	44
Potassium	K	39		19	231	64
Silicon	Si	28		14	118	1410
Sodium	Na	23		11	186	98
Sulfur	S	32		16	102	113

Note: pm = picometre = 10^{-12} m = 1/ 1 000 000 000 000 metres

- Order the elements by weight and write their order numbers into the appropriate column above.
- State at which point (by element name) this order is different to the atomic number order.
- Graph the atomic radius on the vertical axis against the atomic number on the horizontal axis on the first grid on the next page.
- Graph the melting point against the atomic number on the second grid.

The periodic table of elements

5b

Atomic radius (pm)

Atomic number

Melting point (°C)

Atomic number

The periodic table for the first 20 elements

5 Complete the periodic table for the first 20 elements below by writing the symbols into the spaces. Let the atoms with a large atomic radius begin the horizontal periods. Let the elements with a low melting point finish the periods. Let the other elements fit into the periodic table in order of their size and melting point. The first element, hydrogen, is often written above the periodic table, because it has properties from the first and the seventh horizontal group.

						1	2
3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18
19	20						

Trends in the periodic table

Form a diagonal line across the periodic table by shading in the cells of boron and silicon. To the left of this line are the metals, to the right of the line are the non-metals. The shaded elements are semi-metals.

6 Complete the table below by summarising some general properties of metals and non-metals.

Aspect	Metals	Non-metals
Conduct electricity?		
Conduct heat?		
Are solid?		
Are malleable?		

7 Complete the table below by answering 'yes' or 'no' about changes in size, weight and properties of atoms.

Property	Within periods (going across)	Within groups (going down)
Atomic radius increases		
Atomic weight increases		
Atoms have similar properties		

6

 $\oplus$

1



The periodic table of words

7



+

1 Use the clues below to find the down words in the grid.

1	2																	18
		3	4	5	6	7	8	9	10	11	12							

Clues

- 1 Everything that takes up space.
- 2 Element with symbol C.
- 3 A valuable metal used to make jewellery.
- 4 Name of a scientist who concluded that electrons jumped from one energy level to another.
- 5 Dalton's model of an atom looked like a billiard _____.
- 6 Another name for a positive charge: _____ charge.
- 7 Form of energy.
- 8 Planck's first name.
- 9 State of matter where particles are far apart.
- 10 When an atom loses or gains electrons, it becomes an _____.
- 11 Fluorine, chlorine and oxygen are _____-metals.
- 12 Opposite of small.
- 13 State of matter where particles are close to each other.
- 14 Element with the atomic number 18.
- 15 A group of shiny conductors.
- 16 Periodic _____.
- 17 Element with the symbol B.
- 18 Capacity to do work.

2 Once you have written the answers to the clues into the table, the underlined letters become the answer to this question.

What is the name of the group of medieval scientists who wanted to change elements into gold?

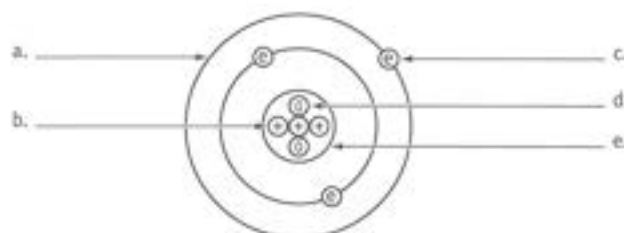


Note: You will need to refer to a copy of the periodic table for this worksheet.

A popular atomic model

Many people today believe that atoms can be described by the atomic model of the solar system atom. This orbital atomic model allows us to explain many processes.

The atom is made up of three main subatomic particles which are grouped into two lots. Protons and neutrons are tightly packed into the nucleus which forms the centre of the atom. Electrons surround the nucleus on several orbits (or shells).



1 Label the diagram of an atomic model (lithium).

Subatomic particles

Protons and neutrons are 2000 times heavier than electrons. Protons and electrons have an electric charge: protons are positive and electrons are negative. Electrons fly around the nucleus on orbits. Neutrons and protons rest in the nucleus.

2 Complete the table about subatomic particles.

Name	Charge	Mass	Position	Movement
			on orbits	
		2000 times		
	neutral			no

Atomic numbers and mass numbers, atoms and isotopes

Atoms are neutral, although some of their subatomic particles do have an electric charge. But the positive and negative subatomic charges cancel each other leaving the atom neutral. Therefore, the number of positive protons must equal the number of negative electrons. The *atomic number* gives the number of protons.

Since protons and neutrons are very much heavier than electrons, it is the protons and neutrons which determine the mass of the atoms. Usually an atom has as many neutrons as protons. The *mass number* is the sum of protons and neutrons.

Some elements have atoms which have the same number of protons in the nucleus (so the same atomic number) but more or fewer neutrons (so they have different atomic weights). Their atoms are now called *isotopes*.

When the number of neutrons is much larger than the number of protons in a large nucleus, the isotope's nucleus is unstable. As a result it breaks up, forming new atoms and giving off radiation.

3 Complete the table about mass and atomic numbers of atoms.

Name	Mass number	Atomic number	Number of protons	Number of neutrons	Number of electrons	Scientific notation
lithium	5	3	3	2	3	${}^5_3\text{Li}$
magnesium	24				12	
aluminium		13		13		
hydrogen	1					
chlorine						${}^{35}_{17}\text{Cl}$
phosphorus						${}^{31}_{15}\text{P}$
oxygen						${}^{16}_8\text{O}$



Electrons on shells

Electrons move on orbits (or shells) around the nucleus. The inside shells are smaller than the outer shells and so can hold fewer electrons. Inner shells are filled up before the outer shells. The first innermost shell can carry a maximum of two electrons, the second and third shells can hold no more than eight electrons (these maximum numbers are true for the first 20 atoms).

- 4** Draw atomic models for the following atoms. The mass and atomic numbers give you a clue to the number of protons, neutrons and electrons.

helium (${}^4_2\text{He}$)

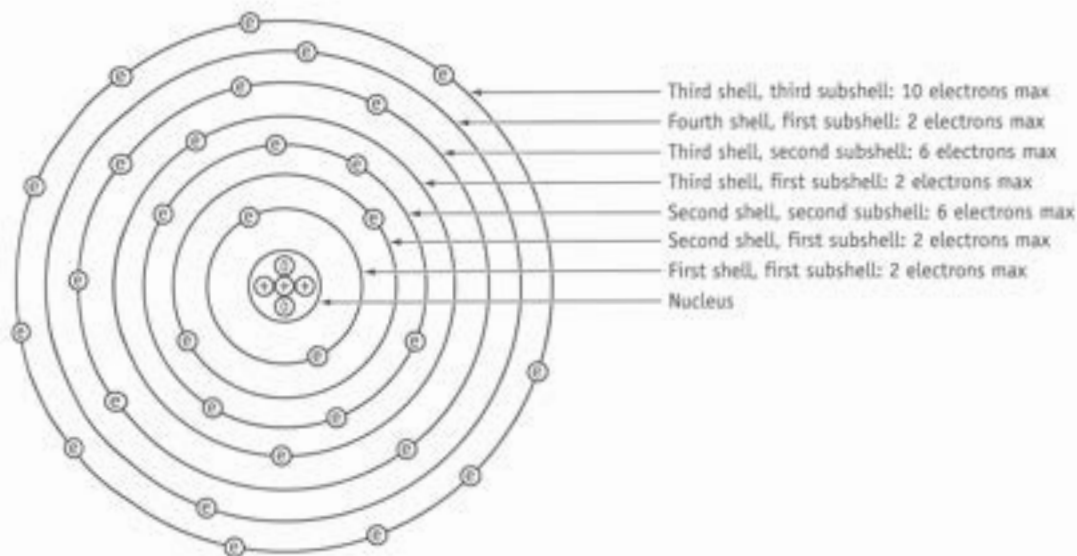
silicon (${}^{28}_{14}\text{Si}$)

beryllium (${}^9_4\text{Be}$)

potassium (${}^{39}_{19}\text{K}$)

Shells and their subshells

The orbital model of the atom becomes untidy with the twenty-first atom, because the electron shells are actually composed of up to four subshells. And some of the subshells overlap each other. The third subshell of the third shell is actually bigger than the first subshell of the fourth shell.



- 5** State the number of protons, neutrons and electrons for the atom of iron (${}^{56}_{26}\text{Fe}$).

- 6** Shade in electrons to the atomic model above for iron.

- 7** State the number of subshells for the first three shells. _____

Around the chemical maze

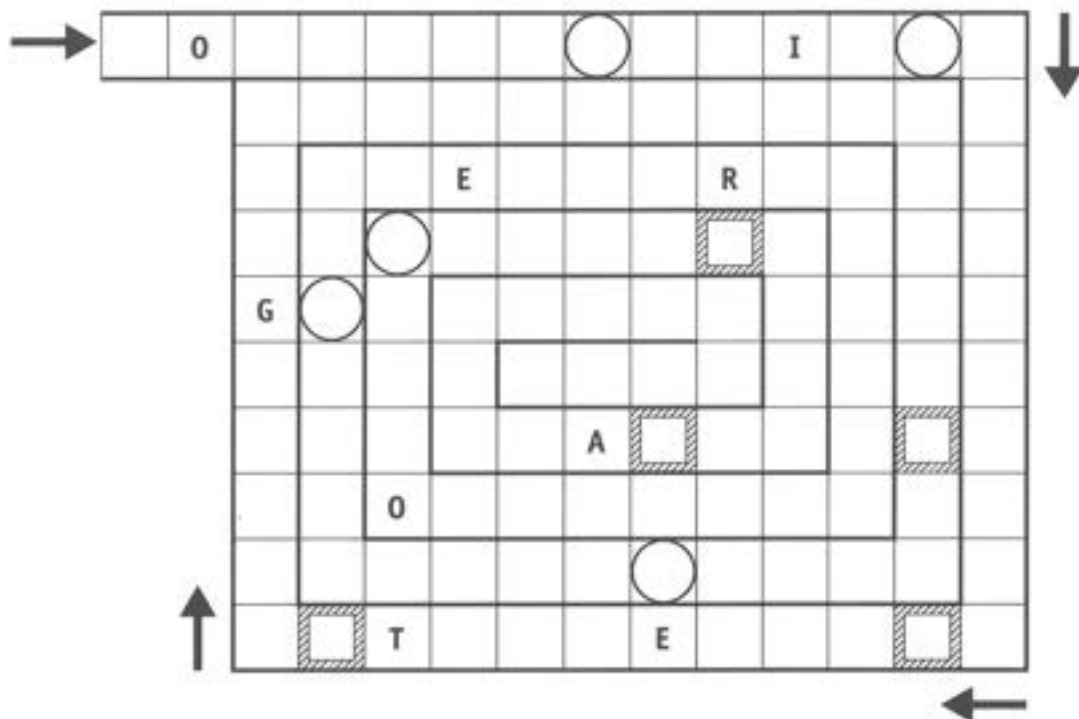
9



Find out who I am by following the instructions below.

In 1566 I was involved in a duel with another student and as a result lost part of my nose. For the rest of my life I wore a metal insert over the missing part. I was involved in the study of heavenly bodies.

Read through the clues and write the answers into the maze, beginning at the arrow at the top left corner. The last letter of each answer is the first letter to the next answer. Some letters are already supplied in the maze. Unjumble the underlined letters to reveal Who I Am.



Clues

- 1 Metal with the symbol Na.
- 2 Shiny malleable substance.
- 3 Element with three protons.
- 4 Able to be pressed or bent into shape.
- 5 Simple pure substance that cannot be broken down into anything simpler.
- 6 Metal with 22 protons.
- 7 Metal with symbol Mg.
- 8 Solids, liquids and gases are all states of _____.
- 9 Noble gas in column 8 of the periodic table.
- 10 Gas in our atmosphere.
- 11 Metal with symbol Ni.
- 12 Water at room temperature.
- 13 Scientist who noticed that lithium, sodium and potassium had very similar properties.
- 14 Dimitri Mendeleev came from which country?
- 15 Metal with symbol Al.
- 16 Scientist who resolved some problems on the periodic table by basing the order on atomic number instead of weight.
- 17 One metal in the periodic table that starts with Y.
- 18 Compound with the formula CH_4 .
- 19 Element with symbol Es.

Who am I?

The circled letters form my given name and the boxed letters my family name. _____

Chemical formulas – what a combo! 10a



+

Note: You will need to refer to a copy of the periodic table for this worksheet.

When you write the formula for a compound, you need to balance its charges by writing numbers as subscripts. For example, Li^+ has a one plus charge and O^{2-} has a two minus charge. When you write a little '2' after the lithium symbol, you are saying that there are two lithiums now. Since each one has a plus one charge, two of them have a two plus charge. And that balances exactly the two minus charge from O^{2-} . Therefore, the formula combining Li^+ and O^{2-} is Li_2O , lithium oxide.



1 Look at the table about chemicals below. Write down each combination in words. Then write down the chemical formula. Some have been done for you already.

	Chloride Cl^-	Oxide O^{2-}	Iodide I^-	Sulfide S^{2-}	Nitrate NO_3^-	Sulfate SO_4^{2-}	Hydroxide OH^-	Carbonate CO_3^{2-}
Lithium Li^+	Lithium chloride LiCl	Lithium oxide Li_2O	Lithium iodide LiI	Lithium sulfide Li_2S	Lithium nitrate LiNO_3	Lithium sulfate Li_2SO_4	Lithium hydroxide LiOH	Lithium carbonate Li_2CO_3
Hydrogen H^+			HI			H_2SO_4		
Magnesium Mg^{2+}			MgI_2		$\text{Mg}(\text{NO}_3)_2$			
Sodium Na^+		Na_2O			NaNO_3			
Copper Cu^{2+}				CuS				CuCO_3
Iron Fe^{3+}	FeCl_3	Fe_2O_3						
Calcium Ca^{2+}				CaS			$\text{Ca}(\text{OH})_2$	
Zinc Zn^{2+}	ZnCl_2					ZnSO_4		
Aluminium Al^{3+}			AlI_3					$\text{Al}_2(\text{CO}_3)_3$

Chemical formulas – what a combo! 10b



+

2 Iron compounds can have two different formulas. For example, another formula for iron chloride is FeCl_2 . This is often called iron II chloride in order to distinguish it from the iron III chloride shown in the table. Write down the formulas of the other iron II compounds from the table.

3 Copper compounds can also have two formulas: copper I compounds and copper II compounds.

a State whether copper I or copper II compounds are in the table. _____

b Write the formulas for the other copper compounds.

4 Use the table above and your periodic table to write the formula for each of these compounds:

a lead nitrate _____

b potassium hydroxide _____

c silver bromide _____

d barium chloride _____

e potassium iodide _____

5 Name the following compounds:

a AgNO_3 _____

b K_2S _____

c BaS _____

d AgI _____

e KI _____

6 Find the following chemical names in the puzzle below. The 13 remaining letters will spell out an important chemical list.

lithium Li hydrogen magnesium Mg sodium Na copper Cu iron Fe calcium
zinc Zn aluminium chloride oxide iodide sulfide nitrate sulfate hydroxide
carbonate lead potassium silver Ag barium bromide

Answer: _____

C	H	L	O	R	I	D	E	Z	S	O	D	I	U	M
A	E	T	A	F	L	U	S	I	M	X	A	R	B	A
L	I	T	H	I	U	M	P	N	G	I	E	O	R	G
C	O	E	Y	R	L	I	D	C	I	D	L	N	O	N
I	D	E	D	I	F	L	U	S	F	E	O	D	M	E
U	I	T	R	A	L	U	M	I	N	I	U	M	I	S
M	D	C	O	P	P	E	R	E	V	L	I	S	D	I
I	E	A	G	M	U	I	S	S	A	T	O	P	E	U
C	U	A	E	H	Y	D	R	O	X	I	D	E	C	M
L	B	G	N	I	T	R	A	T	E	E	N	A	Z	N
B	A	R	I	U	M	E	T	A	N	O	B	R	A	C

Making compounds

11a



+

When atoms lose or gain electrons, they become positive or negative ions. Positive ions join with negative ions to form compounds. In this activity positive ions have a small square sticking out, while negative ions have a small square missing.

hydrogen H^+	hydrogen H^+	chlorine Cl^-	chlorine Cl^-
hydrogen H^+	sodium Na^+	chlorine Cl^-	hydroxide OH^-
sodium Na^+	sodium Na^+	hydroxide OH^-	hydroxide OH^-
magnesium Mg^{2+}	magnesium Mg^{2+}	oxygen O^{2-}	oxygen O^{2-}
calcium Ca^{2+}	calcium Ca^{2+}	oxygen O^{2-}	sulfate SO_4^{2-}
copper Cu^{2+}	zinc Zn^{2+}	sulfate SO_4^{2-}	sulfate SO_4^{2-}
iron Fe^{3+}	iron Fe^{3+}	carbonate CO_3^{2-}	nitrate NO_3^-
			nitrate NO_3^-



- 1 Cut out the positive and negative ions and arrange them to make some compounds. A compound is a large rectangle made up of positive and negative ions, with the small squares sticking out to fit into the missing squares.

Naming compounds

When the formula of a compound is written down, the positive ion is written first and then the negative ion. Subscripts indicate the number of ions involved. For example, HHO becomes H_2O_1 . Note that '1' does not need to be written down, so the correct formula is H_2O .

- 2 Make four compounds and record their formulas by counting the number of ions used.

When writing the name of the compound, only the second part of the name requires attention. The first part of the name is the name of the positive ion (which is often a metal). For the second part, the name of an element changes so that it ends in 'ide'. For example, oxygen becomes oxide. When negative ions ending in 'ate' join with metals, their name stays the same. When they join with hydrogen, the whole name changes. For example, when hydrogen and carbonate join, the new compound is called carbonic acid.

- 3 Construct six compounds and record their formulas and names in the table below.

Compound	Formula	Name
1		
2		
3		
4		
5		
6		

- 4 Make compounds out of the ions specified in the table below. Then write their formulas and names in the table.

Positive ion	Negative ion	Formula	Name
hydrogen	chloride		
hydrogen	sulfate		
hydrogen	hydroxide		
magnesium	sulfate		
magnesium	oxide		
magnesium	hydroxide		
iron III	chloride		
iron III	hydroxide		
iron III	sulfate		
		Fe_2O_3	
		CuSO_4	
		$\text{Cu}(\text{NO}_3)_2$	
		Na_2O	



+ *Note:* You will need to refer to a copy of the periodic table for this worksheet.

Atoms combine with other atoms in order to have complete outermost electron shells. Combined atoms form molecules.

Part 1: Metals combine with non-metals Transferring electrons

1 Look at the periodic table and complete the following sentences.

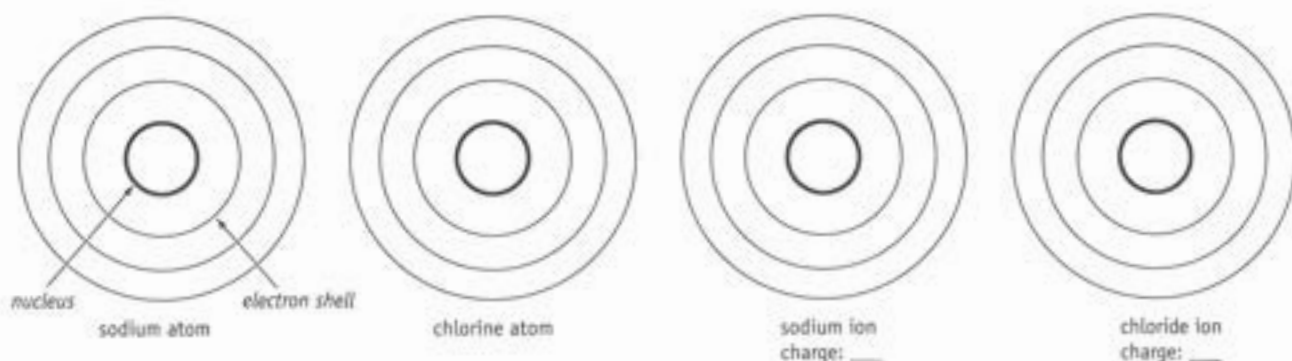
When atoms on the left of the periodic table (ie metals) join with atoms on the right of the periodic table (ie non-metals), electrons are transferred from the metallic atom to the non-metallic atom until the outermost shells are completely full or completely empty.

The metal sodium ($^{23}_{11}\text{Na}$) is in group _____ of the periodic table. On the third electron shell sodium has _____ electron. If it loses that electron, the new outermost shell is the second shell, which has _____ electrons on it, ie it is completely _____. In summary, the sodium atom changed into a positive ion when it lost one _____.

The non-metal chlorine ($^{35}_{17}\text{Cl}$) is in group _____. The outermost electron shell needs to gain _____ more electron to make it completely _____. In summary, the chlorine atom becomes a negative ion, when it _____ an extra electron.

When a sodium atom gives its one lonely electron from the third shell to the chlorine atom, which uses it to fill its outermost shell completely, a molecule has been formed. This molecule, called sodium chloride, consists of two ions: the positive _____ ion and the _____ chlorine ion. The charges of both ions cancel each other, leaving the _____ neutral.

2 a Complete the diagrams of sodium and chlorine atoms by adding the correct number of protons, neutrons and electrons.



b Indicate in the atoms which electron is transferred during the formation of ions.

c Indicate the charges of the ions by writing 1- and 1+.

d Write the name of the resulting molecule: _____

Ions

When atoms lose or gain electrons, they are no longer neutral and are now called ions. Ions form in order to give atoms completely full outermost electron shells.

A metallic atom from the left side of the periodic table usually loses electrons, thus becoming positively charged. A positive ion has more positive protons than negative electrons. If an atom loses two electrons, the resulting ion has a 2+ charge.

A non-metallic atom from the right side of the periodic table usually gains electrons thus becoming negatively charged. A negative ion has extra electrons compared to its atom. If an atom gains three electrons, its resulting ion has a 3- charge.



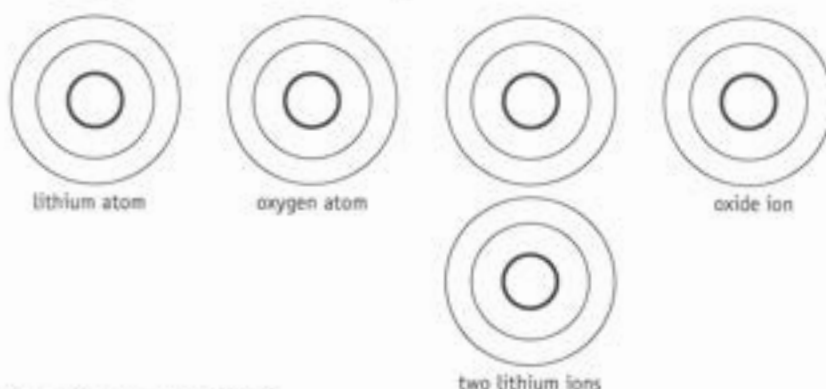
+

- 3** a Name two metals which would form 1+ ions. _____
 b Name two metals which would form 2+ ions. _____
 c Name two non-metals which would form 1- ions. _____
 d Name two non-metals which would form 2- ions. _____
 e Name two non-metals which would form no ions. _____
 f Name two atoms which would form positive or negative ions. _____

Ionic molecules

Sometimes the metallic atom wants to lose only one electron in order to make its new outermost shell complete, but the non-metallic atom requires two electrons to fill its outermost shell completely. In this case, two metallic atoms provide the required electrons. The resulting ionic molecule is made up of two positive ions, each with a 1+ charge, and one negative ion with a 2- charge.

- 4** Complete the atomic diagrams of lithium (${}^7_3\text{Li}$) and oxygen (${}^{16}_8\text{O}$) by adding protons, neutrons and electrons.



The molecule formed from the atoms of lithium and oxygen is called lithium oxide and its formula is Li_2O . The metal is named first and remains unchanged. The non-metal comes last and its ending changes to '-ide'.

- 5** a Name the molecules formed from metal magnesium and non-metal fluorine. _____
 b Name the molecules formed from metal aluminium and non-metal sulfur. _____

It helps to know the charges of the ions in order to determine the number of atoms in a molecule. The lithium ion has a charge of 1+, while the oxygen ion a charge of 2-. The molecule of lithium oxide is made up of 2 lithium ions and 1 oxygen ion. So the charge of the positive ion tells you how many negative ions it combines with.



- 6** Complete the table about ions and molecules.

Molecule	Positive ion	Negative ion	Molecular formula
calcium bromide			
potassium sulfide			
aluminium iodide			
hydrogen oxide			

*Note that hydrogen behaves like a metal here and forms positive ions.

- 7** a State the charge of the complex nitrate ion (NO_3) in the molecule hydrogen nitrate or nitric acid (HNO_3): _____
 b State the charge of the sulfate ion (SO_4) in the molecule hydrogen sulfate or sulfuric acid (H_2SO_4): _____
 c State the charge of the phosphate ion (PO_4) in hydrogen phosphate or phosphoric acid (H_3PO_4): _____

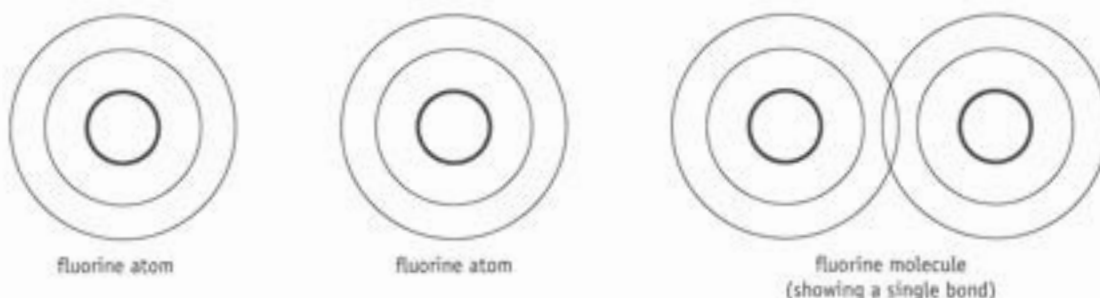


Part 2: Non-metals combine with non-metals

Atoms on the right side of the periodic table (ie non-metals) need only a few electrons to fill their outermost shell completely. When two such atoms want to join to form a molecule, neither one wants to lose electrons and give them to the other. Therefore, ions will not form. When non-metals combine, they share one or two electrons thus ensuring that the outermost shell is completely full.

Fluorine requires one more electron to fill its outermost shell. If it shares one electron with another fluorine atom, its shell is complete. Therefore, two fluorine atoms join to form a fluorine molecule by sharing two electrons, one from each atom. These two electrons make up a single bond. The fluorine molecule can be shown as F-F.

8 Complete the drawings of the fluorine atoms (F_9^{19}) and the fluorine molecule by adding electrons.



When two oxygen atoms join to make an oxygen molecule, they form a double bond. Each atom requires two electrons to make its outermost shell complete. Therefore two single bonds (each made up of two electrons) are required. Two single bonds form one double bond. The oxygen molecule is written in its simplified form as $O=O$.

9 Show the bonds for the molecules of the following non-metals:

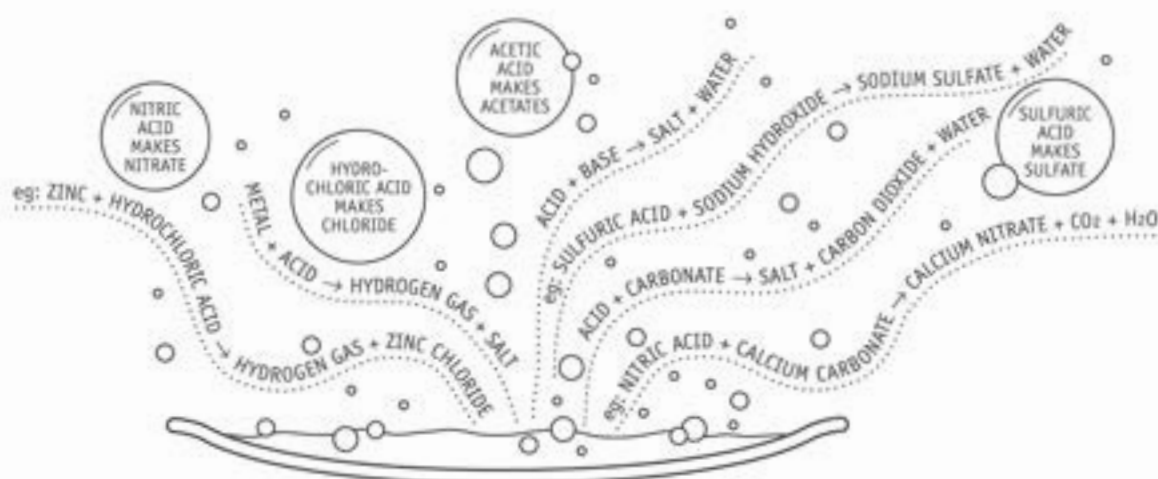
- a chlorine (Cl) _____
- b sulfur (S) _____
- c bromine (Br) _____
- d nitrogen (N) _____

When oxygen (requiring two electrons) and fluorine (requiring one electron) join to form fluorine oxide, two fluorine atoms are needed for each oxygen atom. There is one single bond between the oxygen atom and each fluorine atom resulting in the formula F-O-F or OF_2 .

10 Show the bonds for the following non-metallic molecules:

- a sulfur chloride _____
- b chlorine fluoride _____
- c phosphorus hydride _____
- d nitrogen hydride _____

11 Suggest a reason why helium, neon and argon do not form molecules.



1 Complete the following word equations using the rules above to help you. Then draw a circle around all the acids, box the bases, underline the carbonates and highlight all the metals.

- iron + hydrochloric acid → iron chloride + _____
- magnesium + hydrochloric acid → _____ + hydrogen gas
- zinc + hydrogen sulfate → zinc sulfate + _____
- magnesium + sulfuric acid → _____ + _____
- lead + hydrogen nitrate → _____ + _____
- zinc + acetic acid → zinc acetate + _____
- potassium + acetic acid → _____ + _____
- sodium hydroxide + nitric acid → _____ + _____
- magnesium carbonate + sulfuric acid → _____ + _____ + _____
- calcium carbonate + hydrochloric acid → _____ + _____ + _____
- iron carbonate + acetic acid → _____ + _____ + _____
- _____ + _____ → hydrogen gas + sodium chloride
- _____ + _____ → sodium sulfate + water

2 Write a word equation for the following reactions, then box the reactants and circle the products for each reaction.

- Hydrochloric acid was added to calcium carbonate powder. Bubbles of carbon dioxide formed together with water. The water was evaporated leaving crystals of calcium chloride.

- Sulfuric acid was added to a solution of potassium hydroxide. After the reaction the water that formed was evaporated, leaving behind potassium sulfate.

- A piece of iron was dropped into some nitric acid. Hydrogen gas was produced leaving behind iron nitrate.

- When copper carbonate was added to hydrochloric acid, carbon dioxide and water were produced together with _____.

- A small piece of calcium was added to acetic acid producing calcium acetate and bubbles of _____.



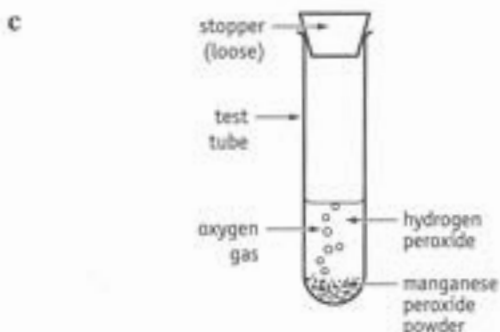
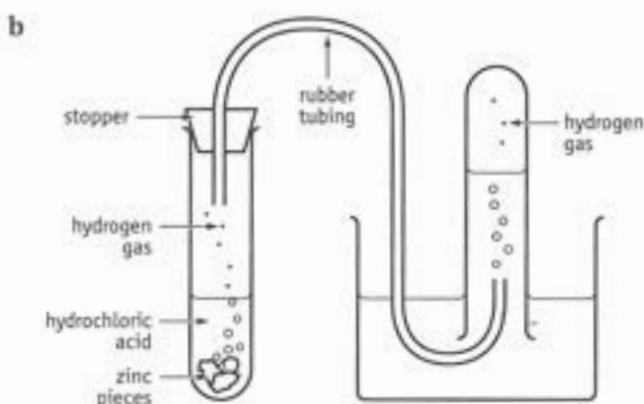
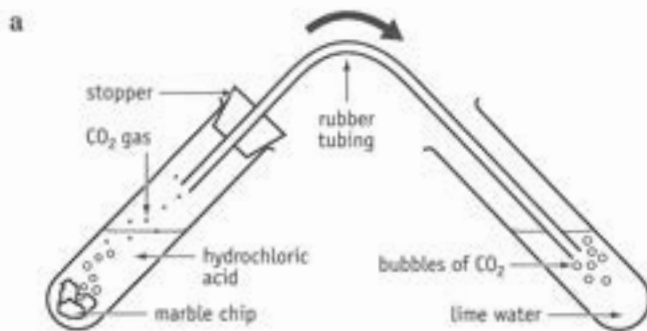
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You may have learnt in the lab how to make different colourless gases and how to test for the presence of each gas. The following activities will look at those experiments about carbon dioxide gas, hydrogen gas and oxygen gas.

Producing the gases

- 1 Carefully observe each diagram below and write a step-by-step procedure on how to conduct each experiment.

Method steps



Testing the gases

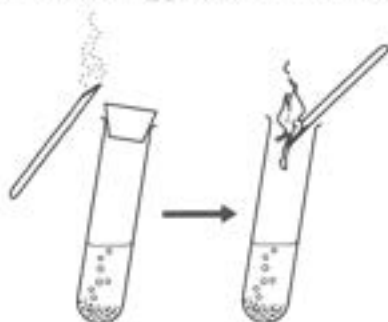
- 2 A student has written some instructions on how to test each of the gases made. Unfortunately the student was a little careless with the instructions and spelling.

Circle the mistakes in each method. There are 10 mistakes in each. Then correct each of the three methods by rewriting them. Use the diagrams to help you.



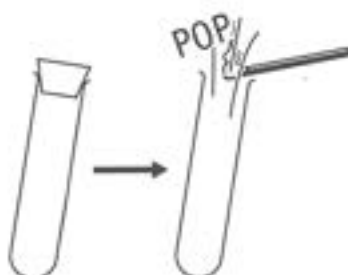
Methods and results

- a ■ Carelessly place a cork inside the test tube with the collected gas.
 ■ Burn a wooden splint, then gently blow it out, making sure the splint is left glowing.
 ■ Throw out the cork and place the glowing splint under the mouth of the test tube.
 ■ Observe what happens or record your results.



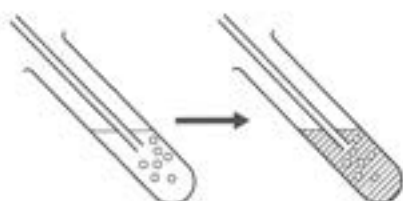
Result: The glowing splint glowed more brightly and flared up.

- b ■ Carefully light a match and hold it till your fingers burn.
 ■ Cork the beaker that contains the collected gas.
 ■ Hold the lit match beneath the test tube.
 ■ Record your observations.



Results: When the gas came in contact with the burning match, it made a 'pop' sound.

- c ■ Collecting the gas produced by bubbling it into some plastic tubing.
 ■ Place the same end of the plastic tubing into some limewater held in a conical flask.
 ■ Make sure the tubing is blocked.
 ■ Throw out the limewater then observe your results.



Results: The clear limewater turned milky.

Corrections



+ Tabulating the gas tests

3 Summarise your results in the table below.

Name of gas	Method for test	Results

Writing word equations

4 Write the word equation for each recounted experiment.

- Carbon dioxide gas is often produced by placing marble chips, which are made of calcium carbonate, in hydrochloric acid. As well as the gas, a salt and water are formed.

- When carbon dioxide is bubbled through limewater, which is calcium hydroxide, water and a white solid form.

- Magnesium added to hydrochloric acid produces a salt and hydrogen gas.

- Hydrogen gas and oxygen from the air explode to produce water vapour.

- When a black powder is added to hydrogen peroxide, the peroxide gives off oxygen gas while the black powder remains unaffected. Water is left when the reaction is completed.

- When the glowing wood flares up, the black charcoal (= carbon) reacts quickly with the oxygen gas to burn into carbon dioxide.

Writing formula equations

5 See whether you can write the reaction equations using the formulas instead of the words.

- Producing carbon dioxide gas: _____
- Performing the gas test with limewater: _____
- Producing hydrogen gas: _____
- Exploding hydrogen gas: _____
- Producing oxygen gas: _____
- Burning away the charcoal: _____



+

When substances burn or combust, they react with oxygen to form *oxides*. Usually oxygen from the air is used, but at times oxygen is supplied from special containers. Extra amounts of oxygen ensure that all substances are completely oxidised.

- 1** Coal is essentially made of carbon. When carbon reacts with oxygen gas from the air, it produces carbon dioxide. Here is the reaction equation in symbols: $C + O_2 \rightarrow CO_2$

Write the reaction equation in words.

- 2** During a science experiment, you held a silver piece of magnesium ribbon into a burner flame, and it suddenly lit up giving off a very bright white light. When the reaction was finished, you were left with a white powder. The magnesium had reacted with oxygen gas.

a Write the word equation and symbol equation for that reaction.

- b When you count the number of oxygen atoms in the reaction above, you should find two oxygen atoms on the left side (O_2 is made up of two O atoms) but only one on the right side (MgO has only one oxygen atom). That means that according to your reaction equation you have lost one oxygen atom during this reaction. However, that is not possible. In order to balance reaction equations, chemists write large numbers in front of the substances until all atoms on the left side of the equation are also on the right side. Let's do this with your equation.

$Mg + O_2 \rightarrow MgO$ *left side: one Mg atom, two O atoms; right side: one Mg atom, one O atom – not balanced*

$Mg + O_2 \rightarrow 2 MgO$ *left side: one Mg atom, two O atoms; right side: two Mg atoms, two O atoms – not balanced*

$2 Mg + O_2 \rightarrow 2 MgO$ *left side: two Mg atoms, two O atoms; right side: two Mg atoms, two O atoms – balanced*

Write the balanced symbol equation for the burning of magnesium.

- 3** The gas that comes out of the gas taps in the laboratory is mainly methane gas, a hydrocarbon.

a State the two elements present in hydrocarbons.

- b When methane burns in oxygen, both elements react with oxygen to produce two oxides. Translate the following symbol equation into a word equation:



- c Balance the symbol equation by writing large numbers in front of some of the substances in the equation. When the equation is balanced, there should be four oxygen atoms and four hydrogen atoms on both sides of the equation.



- 4** When the valve of the Bunsen burner is closed, methane gas mixes with air at the top of the barrel. However, there is not enough oxygen available to oxidise all carbon. That explains the black smudge that appears on the test tube when you heat it with the yellow safety flame.

a Write a word equation and a symbol equation for the incomplete combustion of methane gas.



+

b Write the balanced symbol equation.

c When the valve of the Bunsen burner is opened, complete combustion of methane occurs, ie both elements present in methane are fully oxidised.

Write the word equation for the complete combustion of methane gas.

5 Petrol is also a hydrocarbon. However, it contains 8 carbon atoms and 18 hydrogen atoms and it is called *octane*. When a car engine is working properly, all carbon and hydrogen is oxidised.

a Write the word equation and symbol equation for the complete combustion of octane.

b Write the balanced symbol equation.

c When a car engine is not working properly, poisonous carbon monoxide gas and black carbon powder, as well as carbon dioxide and water, are produced.

Write the word equation and symbol equation for the incomplete combustion of petrol.

d Write the balanced symbol equation for the incomplete combustion of octane. (Hint: Balance the carbon first, then the hydrogen.)

e Suggest how the complete combustion of petrol could be ensured in an engine.

6 White table sugar is a carbohydrate called sucrose. Sucrose is made up of many glucose molecules. The formula of glucose is $C_6H_{12}O_6$.

a Name the elements present in a carbohydrate.

b The cells in your body receive energy during the process called respiration. In respiration glucose reacts with oxygen gas.

Write the word equation and a symbol equation for respiration.

c Write a balanced symbol equation for respiration.

d Explain how glucose and oxygen are transported to muscle cells in your body.

e Explain how your body gets rid of the two waste products of respiration.

f When your body has to work very hard, for example during a strenuous sports event, there is not enough oxygen available for the complete oxidation of glucose. As a result, the two waste products are not produced and taken away. Instead other chemicals are produced which stay in your muscles and cause muscular pain. The main chemical which gives you sore muscles is called lactic acid. Suggest how you could prevent or ease sore muscles.



Acid rain is a major problem in countries that are industrialised. Such countries tend to release large amounts of sulfur dioxide and nitrogen dioxide into the air. The former gas is emitted into the air by burning coal, which contains some sulfur, and by extracting metals from sulfide ores. The latter gas comes from the use of petrol and diesel in cars and trucks.

Nitrogen dioxide gas dissolves in rainwater drops to produce nitric acid. Sulfur dioxide in the atmosphere also dissolves in water droplets in the air to produce sulfuric acid. As a result of those acids in the rain drops, acid rain is formed.

Acid rain has a low pH, as low as 3. It is therefore quite acidic. The long-term effects of this rain have unfortunately lowered the pH of rivers and lakes. This has had a detrimental effect upon fish populations. Plants also suffer because the soil becomes affected as well. Such effects have been seen in North America and Scandinavia. In Europe, acid rain has done more damage to statues and old buildings than weathering has done in the last 400 years.

Looking at the text

- 1 Write down the meanings of the six underlined words in the text.

- 2 Write the main theme of each paragraph.

a Paragraph 1: _____

b Paragraph 2: _____

c Paragraph 3: _____

- 3 State the purpose of this text. _____

- 4 Pronouns are words such as *he*, *she*, *it* and *they*, which are used to refer to nouns that have been used earlier in the text. Find two pronouns in the text above and circle them. Then write down the pronouns and the noun they are referring to.

- 5 Another small but important part of text is the pointing words (also called determiners or articles). When you are asked to test for *a gas*, you may not know which gas you will find. But once you have finished testing, you know exactly which gas you are talking about, and from then on can refer to *the gas*. Therefore, using words like *a*, or no pointing word at all, keeps things very general, but using pointing words like *the* makes things very specific.

Find four specific pointing words in the text above and box them. Then write down the term with the pointing word and the specific term they are referring to.



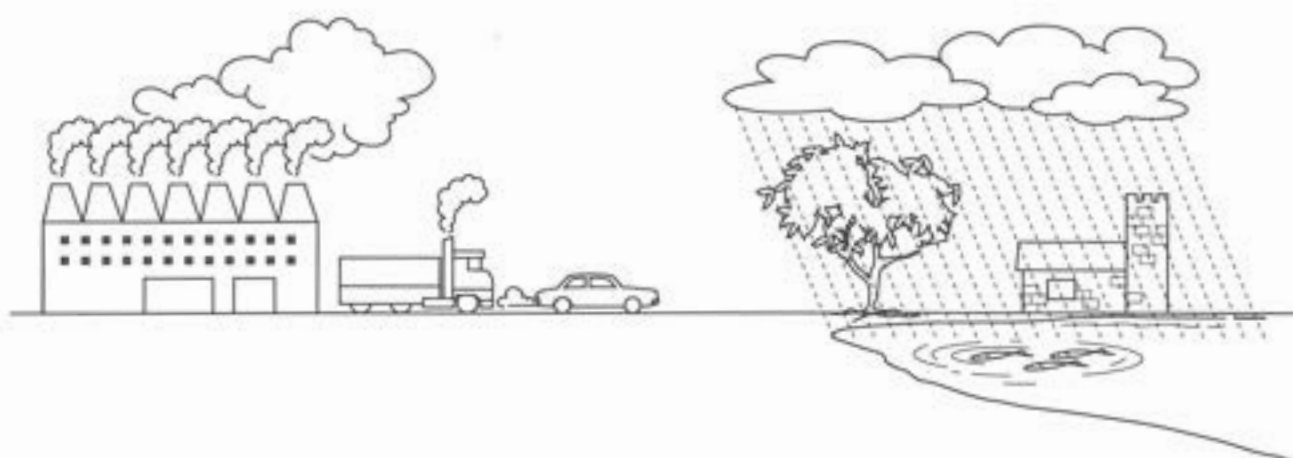
Learning from the text

- 6 Suggest what effect acid rain would have on metals such as iron by writing word equations for the reactions. _____
- 7 Mortar usually contains limestone (CaCO_3). Write a word equation to show the effect acid rain would have mortar. _____
- 8 Explain why Australia is not strongly affected by acid rain. _____

Labelling a diagram

- 9 Cut out the following labels and paste them in the appropriate places on the diagram below. Use the text above to help you.

1 Acids destroy trees and damage the soil.	2 Sulfur dioxide is emitted.	3 When nitrogen combusts in car engines, nitrogen dioxide is formed.
4 Sulfur dioxide and rain water combine to form sulfuric acid.	5 Nitrogen dioxide is given out.	6 When fossil fuels are burnt in factories, sulfur is changed into sulfur dioxide.
7 Combining nitrogen dioxide with rain water results in nitric acid.	8 Acids attack buildings and statues.	9 Acid rain lowers the pH of rivers and lakes and so reduces fish populations.



An acid rain recount mix up

18a



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The following recount is an experiment done by a science class on acid rain.

1 Read the sections carefully, cut them out and arrange the recount sentences in the correct order.

a	First we added a piece of blue litmus paper to one of the test tubes labelled A. It turned red.
b	For each of the test tubes labelled A we performed the following tests.
c	Lastly we added a few drops of universal indicator to the fourth test tube labelled W. The colour remained green.
d	Lastly we added a few drops of universal indicator to the last test tube labelled A. The test tube turned red.
e	Our job was to find out how acid rain and plain water affect various things.
f	Our teacher mixed the sulfur dioxide gas with water to represent acid rain.
g	Yesterday our teacher demonstrated the burning of some sulfur on a spoon in the fume cupboard.
h	Second we added three small pieces of marble chips (= calcium carbonate). We noticed fizzing.
i	The burning of sulfur on the spoon represented coal which was burnt in a factory and produced sulfur dioxide gas.
j	Then we added magnesium ribbon to the third test tube A. This also fizzed.
k	There was no reaction at all when we added magnesium ribbon and then marble chip to a test tube W.
l	We concluded that the sulfur dioxide has an effect on water and that the water then behaves like an acid.
m	We labelled the four test tubes with A for acid rain and the other four with W for water.
n	We repeated the experiment with the four test tubes labelled W.
o	We were given four test tubes with acid rain and four test tubes with water.
p	When we tested a test tube W with blue litmus paper, it stayed blue.

2 After you have sequenced the sentences, change the recount into a procedure. Fill in the result table using the students' results and finish the experiment with a conclusion.

Title	
Aim	
Equipment	
Method	

+

Labelled diagram

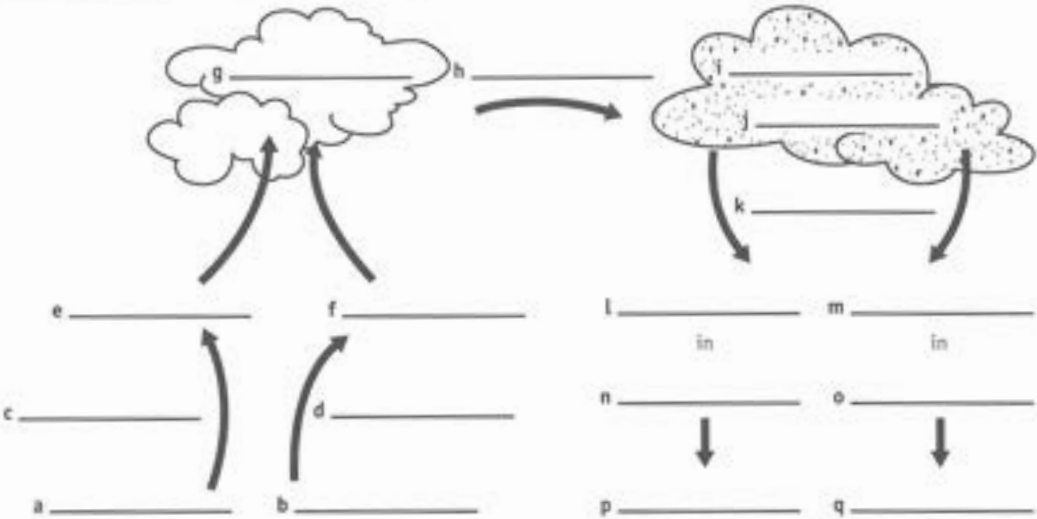
Result

	Acid rain water	Pure water
Blue litmus		
Marble chips		
Magnesium ribbon		
Universal indicator		

Conclusion

3 a The following flow chart represents a typical acid rain cycle. Complete the flow chart by writing the missing labels into the appropriate lines. Choose the terms from the box.

acid rain buildings burning fossils fuels cars, trucks collapsed buildings combusting petrol
dead fish destruction of mortar dissolution factories increase in acidity rain droplets
nitric acid rivers, lakes sulfur dioxide sulfuric acid nitrogen dioxide



- b Draw a box around each object (eg factory).
- c Draw a circle around each process (eg burning).
- d Draw a line under each product (eg sulfur dioxide).

The useful game of acids, bases and salts 19a



Note: You will need one numbered die to play this game.

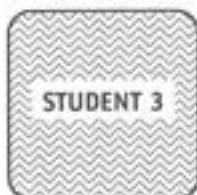
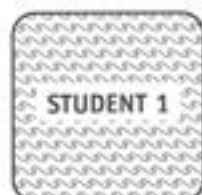
Cut out the FREE TURN tokens and the markers (named STUDENT 1 etc) below.

The aim of the game is to move a marker across the game in order to collect information, which is then recorded into a table about acids, bases and salt. The person who completes the table first is the winner.

Rules

- You can play in pairs or groups of three or four players.
- Roll the die.
- Begin at any corner move your marker horizontally or vertically (not diagonally) for the number of spaces indicated by the die.
- If you land on an acid formula square, move back two spaces; if you land on a base formula square, move back three spaces; and if you land on a salt formula square, miss a turn.
- If you end up on any square with a formula, say the name of the compound correctly and score a free turn. Take a FREE TURN token. You can use the free turn any time when it is your turn.
- When you end up on a square with text, record the information in the table below.
- The winner of the game is the player who has completed the table first.

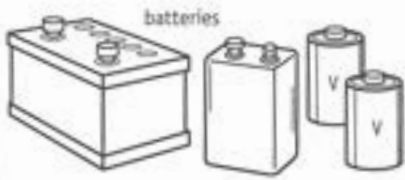
Chemical	Category	Use
	Acid	
	Acid	
	Acid	
	Acid	
	Acid	
	Base	
	Base	
	Base	
	Base	
	Base	
	Salt	
	Salt	
	Salt	
	Salt	
	Salt	



The useful game of acids, bases and salts

19b



Antacid tablets contain calcium carbonate which neutralises hydrochloric acid in the stomach.	Acetic acid is used as flavouring and preservative in foods.	HCl	Ammonium chloride is one of the constituents of dry cell batteries.	Window cleaners contain a weak base for dissolving dirt on windows.	Tartaric acid is used in cooking. It reacts to produce carbon dioxide causing cakes to rise.
Potassium hydroxide is used to make soap and helps to clean our skin.	NaCl			Ba ₂ SO ₄	Caustic soda (= sodium hydroxide) strips paint off wood.
Bluestone (= copper sulfate) is used as a fungicide in swimming pools and on trees.	Ammonia is a weak base in cleaning agents for dirty floors.	Na ₂ CO ₃	Car batteries contain sulfuric acid.	Fertilisers are manufactured with sulfuric acid. They are also used to make petrol, kerosene and other fuels.	NaOH
Toothpastes are slightly basic to neutralise the acids produced by decaying food.	KOH	Nitric acid is used in the manufacture of fertilisers.	HNO ₃	Sodium hydroxide is used to clean drains, ovens and surfaces which are very dirty and greasy.	Shampoos contain a weak base to clean our hair.
H ₂ SO ₄		Hydrofluoric acid is used for etching glass. It is one of the strongest acids.			Hydrochloric acid is used to clean mortar containing calcium carbonate from bricks.
Marble (= calcium carbonate) is used as ornamental stone.		MgCl ₂	Acetic acid is used to make clothing, paint, medicines, rubber and glass.	Calamine lotion contains a weak base to relieve insect bites that are acidic.	Plaster of Paris (= calcium sulfate) is used for plastering walls and for making casts to hold broken limbs still.
Sodium carbonate is used in the manufacture of glass.	Ba(OH) ₂	Carbonic acid puts the fizz in soft drinks.	White lead or lead carbonate is in some paints.	CaCO ₃	Hydrochloric acid is used to etch glass.

Indicators

Acids and bases change the colour of certain dyes. These dyes are called indicators. Indicators tell us whether a substance is acidic, basic or neutral. Some indicators in the school laboratory are kept as liquids in dropper bottles, while others are dried onto strips of paper. Litmus, phenolphthalein, methyl orange and bromothymol are some common indicators. Indicators can also be made from plants, for example, the liquid from boiled rhubarb stems turns red in acids and green in bases.

pH scale

Scientists have an accurate way of measuring acidity. They use a very sensitive indicator called pH paper. The pH paper changes into one of several colours when dipped into a liquid substance. The colour is matched to numbers on a pH scale. The pH scale gives very acidic solutions the number 1, very basic solutions number 14, and neutral solutions number 7.

Diagram 1: Some substances and their pH numbers

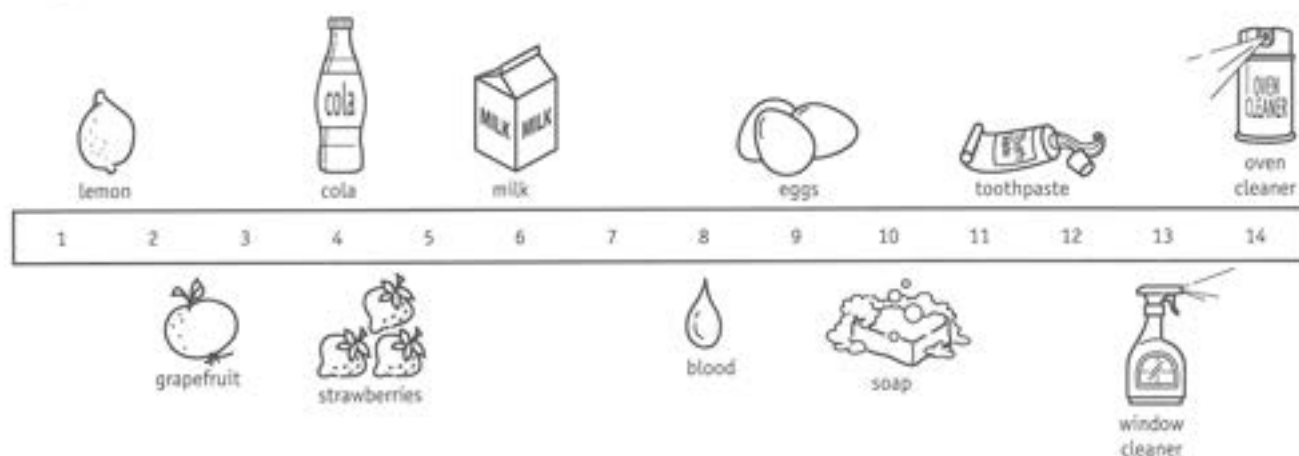


Diagram 2: Indicator colours and the pH scale

pH	1	2	3	4	5	6	7	8	9	10	11	12	13	14
	strongly acidic			weakly acidic		neutral		weakly basic		strongly basic				
litmus				red						blue				
methyl orange	red								yellow					
pH paper		red			yellow		pale yellow			green			blue	
phenolphthalein					colourless			pink				red-purple		
thymol blue		red				yellow			green			blue		
universal	red			orange		yellow	green			blue		purple	deep purple	

Table 1: Plant indicators and colours

Plant source	Colour of indicator solution		
	as extracted	in acids	in bases
hydrangea	colourless	pink	yellow
pansy	purple	pink	yellow
pink geranium	pink	pink	orange
red cabbage	purple	red	green
rhubarb	red	red	pale green
rose (pink)	yellow	pink	orange
sweet pea	light green	pink	yellow



+

Use the text, diagrams and table above to answer the following questions.

- 1** State the colour of universal indicator for the common substances in diagram 1. Record your answers in the table below.

Substance	Colour of universal indicator

- 2** State the colour of litmus if the plant indicators in Table 1 are either orange, red or yellow. Tabulate your results.

- 3** Five unknown solutions were tested with indicators by a group of students. Their results are shown below.

Solution	Indicator	Colour
A	pH paper	pale yellow
B	methyl orange	red
C	universal	purple
D	thymol blue	green
E	litmus	blue

- a Name the acidic solutions. _____
 b List the basic solutions. _____
 c State the neutral solutions. _____

- 4** The students were then to test each solution A, B, C, D and E with the plant indicators red cabbage, rhubarb and rose. Show what their results should be in the table below.



+

- 5** The teacher provided the group of students with an unknown solution F. The students' task was to find the pH of the solution. They tested the solution with several indicators as shown in their results table.

Indicator	Colour
hydrangea	pink
methyl orange	yellow
phenolphthalein	colourless
thymol blue	yellow

Deduce the pH range of solution F. _____

- 6** a List the solutions A, B, C, D and F in order of their acidity.

Most acidic _____ Most basic _____

- b Explain why the solution E was not included in the question above.

- 7** a Explain why labels on cleaning substances for ovens and drains warn the user to wear gloves and not to swallow the substances.

- b Describe what you would do if some cleaning substance splashed into your eye.

- 8** A gardener measures some garden soil and finds that it has a pH of 5.2. Since the flowers planted there grow best at a pH of 7, the gardener adds some crushed limestone to the soil. Explain whether limestone is acidic, neutral or basic.

- 9** Using your knowledge of the pH scale complete the table below, listing any foods, drinks and household products you use regularly in the correction columns.

Acidic	Neutral	Basic

Cheminoes – elements and compounds 21



Below is a domino game with the names and symbols of common elements and compounds.

Instructions:

- Cut out the chemino pieces by cutting along the thick lines.
- Place them face down on the table.
- Divide the pieces between each of two to four players.
- One piece is placed face up in the middle of the table. Each player in turn joins one piece onto either end of the pieces on the table, making a long chain. The symbol and name of the same substance are to be matched.
- The first person to use all their pieces is the winner.

sodium chloride	copper	hydrochloric acid	oxygen	hydrogen	carbon dioxide gas	carbon monoxide	hydrochloric acid
MgO	O ₂	Mg	CO	CuSO ₄	ZnCl ₂	Cu	H ₂ O
calcium	oxygen gas	nitric acid	helium gas	sulfur dioxide	zinc chloride	sodium	water
NaOH	N	Zn	CO ₂	HNO ₃	SO ₂	S	NaCl
nitrogen	magnesium	sodium hydroxide	chlorine gas	sodium	nitrogen dioxide	sodium hydroxide	table salt
H ₂ SO ₄	H ₂ O	Ca	K	NO ₂	MgCl ₂	Cl ₂	HCl
sulfuric acid	sulfur	copper sulfate	iron	potassium	water	oxygen gas	carbonic acid
H ₂	H	Na	He	CuSO ₄	Fe ₂ O ₃	NaOH	CO ₂
iron oxide	hydrogen gas	calcium carbonate	chlorine	carbon	magnesium oxide	magnesium chloride	carbon dioxide gas
HCl	NaCl	Cl	Fe	CO	CaCO ₃	O ₂	H ₂ CO ₃



+

Below is a domino game about the chemical and common names of compounds.

Instructions:

- Cut out the chemino pieces by cutting along the thick lines.
- Place them face down on the table.
- Divide the pieces between each of two to four players.
- One piece is placed face up in the middle of the table. Each player in turn joins one piece onto either end of the pieces on the table, making a long chain. The chemino pieces show the chemical symbol and its common name. The symbol is to be matched with its common name.
- The first person to use all their pieces is the winner.

The following table will help you to translate the symbol into its common name.

Chemical name	Common name
Ammonia	Nitrogen trihydride
Bicarbonate soda	Sodium hydrogen carbonate
Bluestone	Copper sulfate
Brimstone	Sulfur
Caustic potash	Potassium hydroxide
Caustic soda	Sodium hydroxide
Chalk	Calcium carbonate
Chloroform	Trichloro-methane
Condy's crystals	Potassium permanganate
Dry ice	Solid carbon dioxide
Epsom salts	Magnesium sulfate
Laughing gas	Dinitrogen monoxide
Lime	Calcium oxide

Chemical name	Common name
Limestone	Calcium carbonate
Magnesia	Magnesium
Muriatic acid	Hydrochloric acid
Peroxide	Hydrogen peroxide
Potash	Calcium carbonate
Quicklime	Calcium oxide
Quicksilver	Mercury
Rust	Iron oxide
Saltpetre	Potassium nitrate
Table salt	Sodium chloride
Washing soda	Sodium carbonate
Water	Dihydrogen monoxide

NH ₃	peroxide	H ₂ O	washing soda	Fe ₂ O ₃	potash	solid CO ₂	laughing gas	KOH	brimstone
S	caustic soda	Hg	chalk	CaCO ₃	Epsom salt	NaCl	quicksilver	N ₂ O	lime
HCl	magnesia	NaOH	quicklime	CaO	rust	Na ₂ CO ₃	table salt	CaO	limestone
CuSO ₄	caustic potash	CHCl ₃	Condy's crystals	KMnO ₄	saltpetre	H ₂ O	muriatic acid	MgSO ₄	dry ice
NaHCO ₃	water	H ₂ O ₂	Glauber's salt	Na ₂ SO ₄	chloroform	K ₂ CO ₃	bluestone	KNO ₃	ammonia
H ₂ O ₂	bicarbonate soda	NaOH	dry ice	MgSO ₄	table salt	HCl	caustic soda	H ₂ O	lime



1 History of the atomic model

- 1 to explain, perform, show, looks, to develop, has undergone, was suggested, deduced, had to be, called, heard, concluded, were, could not be, was, experimented, found, were, travelled, carried, came, were, concluded, were made up of, were, had, came, had to, contained
- 2 particles, through, unaffected, surprised, source, surround, field, quantum, solar, likely, appeared, existence
- 3 a Democritus from ancient Greece named the smallest particle atom in about 400 BC.
b The English scientist John Dalton believed that the atom was like a ball in 1803.
c In 1904 Joseph Thomson from England drew the atom like a positively charged plum pudding with negative plums in it.
d In 1911 Ernest Rutherford, who was born in New Zealand, concentrated the positive charge in the central nucleus and moved the negative electrons onto a cloud surrounding the nucleus.
e The Danish Niels Bohr placed the negative electrons onto several orbits rather like our solar system but kept the positive nucleus in 1913.
f The Austrian scientists Erwin Schrodinger and Wolfgang Pauli combined the idea of orbits and clouds in 1926 and developed an atom where the nucleus was surrounded by several orbital clouds.
g In 1932 the English scientist James Chadwick discovered neutral neutrons inside the positive nucleus.

2 Amazing atoms

Start – Planck – protons – Rutherford – Germany – positive terminal – Bohr – negative terminal – electricity – English – Dalton – Rutherford – Schrodinger and Pauli – electrons – Thomson – Greece – Bohr

3 Atoms and molecules

1	Particles	Atoms	Molecules
	Diagrams	A, I, K, O	B, C, D, E, F, G, H, J, L, M, N

2	Substances	Elements	Compounds
	Diagrams	A, F, H, I, J, K, N, O	B, C, D, E, G, L, M

3	Atom of element	Atom of compound	Molecule of element	Molecule of compound
	A, I, K, O	—	F, H, J, N	B, C, D, E, G, L, M

- 4 You cannot have atoms (single parts) of compounds (two or more different parts).
- 5 Teacher to check.
- 6 The lines are the chemical bonds holding atoms together in a molecule.

4 Eureka! Solve this puzzle

- 1 matter, atoms, substance; molecule, particles, solid, bonds, form, criteria; metals, liquid, solid; gas, compounds, water, silver, hydrogen, element
King's problem: Did the goldsmiths use all the gold or replace some of it with silver when making the golden crown?
- 2 Teacher to check.
- 3 Archimedes solved the problem in the bath tub.

5 The periodic table of elements

- 1 In order down the column: 13, 19, 4, 5, 20, 6, 17, 9, 2, 1, 3, 12, 10, 7, 8, 15, 18, 14, 11, 16
- 2 Ar, C, N, K
- 3–5 Teacher to check.

6	Aspect	Metals	Non-metals
	Conduct electricity?	yes	no
	Conduct heat?	yes	poorly/no
	Are solid?	yes (except Mercury)	some
	Are malleable?	yes	no

7	Property	Within periods (going across)	Within groups (going down)
	Atomic radius increases	no (slowly decreases)	yes
	Atomic weight increases	yes (slowly)	yes
	Atoms have similar properties	no	yes



6 What a wheely atomic jumble!

- 1 What is the name of the metal with the symbol Na? Sodium
- 2 What is the positive particle found in the nucleus called? Proton
- 3 What is the name of the liquid metal at room temperature? Mercury
- 4 What are the negative particles outside the nucleus called? Electron
- 5 What is the name of the table that contains all the elements? Periodic
- 6 What do we call anything which takes up space? Matter

7 The periodic table of words

1. matter 2. carbon 3. gold 4. Bohr 5. ball 6. plus 7. heat 8. Max 9. gas 10. ion 11. non 12. big 13. solid 14. Argon 15. metal 16. table 17. Boron 18. energy
- 2 alchemists

8 Atomic structure

- 1 a orbit b proton c electron d neutron e nucleus

2

Name	Charge	Mass	Position	Movement
electron	negative	1	on orbits	yes
proton	positive	2000 times	in nucleus	no
neutron	neutral	2000 times	in nucleus	no

3

Name	Mass number	Atomic number	Number of protons	Number of neutrons	Number of electrons	Scientific notation
lithium	5	3	3	2	3	${}^5_3\text{Li}$
magnesium	24	12	12	12	12	${}^{24}_{12}\text{Mg}$
aluminium	26	13	13	13	13	${}^{26}_{13}\text{Al}$
hydrogen	1	1	1	0	1	${}^1_1\text{H}$
chlorine	35	17	17	18	17	${}^{35}_{17}\text{Cl}$
phosphorus	31	15	15	16	15	${}^{31}_{15}\text{P}$
oxygen	16	8	8	8	8	${}^{16}_8\text{O}$

5 26, 30, 26

7 1: 1, 2: 2, 3: 3

9 Around the chemical maze

sodium, metal, lithium, malleable, element, titanium, magnesium, matter, radon, nitrogen, nickel, liquid, Dobereiner, Russia, aluminium, Mosely, ytterbium, methane, Einsteinium

Who Am I? Tycho Brahe

10 Chemical formulas – what a combo!

1

	Chloride Cl^-	Oxide O^{2-}	Iodide I^-	Sulfide S^{2-}	Nitrate NO_3^-	Sulfate SO_4^{2-}	Hydroxide OH^-	Carbonate CO_3^{2-}
Lithium Li^+	Lithium chloride LiCl	Lithium oxide Li_2O	Lithium iodide LiI	Lithium sulfide Li_2S	Lithium nitrate LiNO_3	Lithium sulfate Li_2SO_4	Lithium hydroxide LiOH	Lithium carbonate Li_2CO_3
Hydrogen H^+	Hydrogen chloride HCl	Hydrogen oxide H_2O	Hydrogen iodide HI	Hydrogen sulfide H_2S	Hydrogen nitrate HNO_3	Hydrogen sulfate H_2SO_4	Hydrogen hydroxide H_2O	Hydrogen carbonate H_2CO_3
Magnesium Mg^{2+}	Magnesium chloride MgCl_2	Magnesium oxide MgO	Magnesium iodide MgI_2	Magnesium sulfide MgS	Magnesium nitrate $\text{Mg}(\text{NO}_3)_2$	Magnesium sulfate MgSO_4	Magnesium hydroxide $\text{Mg}(\text{OH})_2$	Magnesium carbonate MgCO_3
Sodium Na^+	Sodium chloride NaCl	Sodium oxide Na_2O	Sodium iodide NaI	Sodium sulfide Na_2S	Sodium nitrate NaNO_3	Sodium sulfate Na_2SO_4	Sodium hydroxide NaOH	Sodium carbonate Na_2CO_3
Copper Cu^{2+}	Copper chloride CuCl_2	Copper oxide CuO	Copper iodide CuI_2	Copper sulfide CuS	Copper nitrate $\text{Cu}(\text{NO}_3)_2$	Copper sulfate CuSO_4	Copper hydroxide $\text{Cu}(\text{OH})_2$	Copper carbonate CuCO_3
Iron Fe^{2+}	Iron chloride FeCl_2	Iron oxide Fe_2O_3	Iron iodide FeI_2	Iron sulfide FeS	Iron nitrate $\text{Fe}(\text{NO}_3)_3$	Iron sulfate $\text{Fe}_2(\text{SO}_4)_3$	Iron hydroxide $\text{Fe}(\text{OH})_3$	Iron carbonate $\text{Fe}_2(\text{CO}_3)_3$

Calcium Ca^{2+}	Calcium chloride CaCl_2	Calcium oxide CaO	Calcium iodide CaI_2	Calcium sulfide CaS	Calcium nitrate $\text{Ca}(\text{NO}_3)_2$	Calcium sulfate CaSO_4	Calcium hydroxide $\text{Ca}(\text{OH})_2$	Calcium carbonate CaCO_3
Zinc Zn^{2+}	Zinc chloride ZnCl_2	Zinc oxide ZnO	Zinc iodide ZnI_2	Zinc sulfide ZnS	Zinc nitrate $\text{Zn}(\text{NO}_3)_2$	Zinc sulfate ZnSO_4	Zinc hydroxide $\text{Zn}(\text{OH})_2$	Zinc carbonate ZnCO_3
Aluminium Al^{3+}	Aluminium chloride AlCl_3	Aluminium oxide Al_2O_3	Aluminium iodide AlI_3	Aluminium sulfide Al_2S_3	Aluminium nitrate $\text{Al}(\text{NO}_3)_3$	Aluminium sulfate $\text{Al}_2(\text{SO}_4)_3$	Aluminium hydroxide $\text{Al}(\text{OH})_3$	Aluminium carbonate $\text{Al}_2(\text{CO}_3)_3$

2 FeCl_3 FeO FeI_2 FeS $\text{Fe}(\text{NO}_3)_2$ FeSO_4 $\text{Fe}(\text{OH})_2$ FeCO_3

3 a copper II compounds

b CuCl Cu_2O CuI Cu_2S CuNO_3 Cu_2SO_4 CuOH Cu_2CO_3

4 a $\text{Pb}(\text{NO}_3)_2$ b KOH c AgBr d BaCl_2 e KI

5 a silver nitrate b potassium sulfide c barium sulfide

d silver iodide e potassium iodide

6 Periodic table

11 Making compounds

3 eg

Compound	Formula	Name
1	HCl	Hydrogen chloride (or hydrochloric acid)
2	MgCl_2	Magnesium chloride
3	$\text{Ca}(\text{OH})_2$	Calcium hydroxide
4	ZnO	Zinc oxide
5	FeCl_3	Zinc chloride
6	$\text{Fe}_2(\text{SO}_4)_3$	Iron sulfate

Positive ion	Negative ion	Formula	Name
hydrogen	chloride	HCl	Hydrogen chloride (or hydrochloric acid)
hydrogen	sulfate	H_2SO_4	Hydrogen sulfate (or sulfuric acid)
hydrogen	hydroxide	HOH or H_2O	Hydrogen hydroxide (or water)
magnesium	sulfate	MgSO_4	Magnesium sulfate
magnesium	oxide	MgO	Magnesium oxide
magnesium	hydroxide	$\text{Mg}(\text{OH})_2$	Magnesium hydroxide
Iron III	chloride	FeCl_3	Iron chloride
Iron III	hydroxide	$\text{Fe}(\text{OH})_3$	Iron hydroxide
Iron III	sulfate	$\text{Fe}_2(\text{SO}_4)_3$	Iron sulfate
Iron III	oxide	Fe_2O_3	Iron oxide
copper	sulfate	CuSO_4	Copper sulfate
copper	nitrate	$\text{Cu}(\text{NO}_3)_2$	Copper nitrate
sodium	oxide	Na_2O	Sodium oxide

12 Chemical formulas

1 1, 1, 8, full, electron; 7, 1, full, gains; sodium, negative, molecule

2 d sodium chloride

3 a eg lithium, sodium, potassium, rubidium

b eg beryllium, magnesium, calcium, strontium

c eg fluorine, chlorine, bromine, iodine

d eg oxygen, sulfur, selenium

e eg helium, neon, argon, krypton

f eg carbon, silicon

4 Teacher to check.

5 a magnesium fluoride b aluminium sulfide



+

Molecule	Positive ion	Negative ion	Molecular formula
calcium bromide	calcium (2+)	bromide (1-)	CaBr ₂
potassium sulfide	potassium (1+)	sulfide (2-)	K ₂ S
aluminium iodide	aluminium (3+)	iodide (1-)	AlI ₃
hydrogen oxide	hydrogen (1+)	oxide (2-)	H ₂ O

7 a 1- b 2- c 3-

8 Teacher to check.

9 a Cl - Cl b S = S c Br - Br d N = N

10 a Cl - S - Cl, SCl₂ b Cl - F
c H - P - H - H, PH₃ d H - N - H - H, NH₃

11 They have no need to share or accept electrons to gain a complete outermost shell, as their outermost shell is already full.

13 Chemical word equations

- 1 a hydrogen gas b magnesium chloride
c hydrogen gas d magnesium sulfate + hydrogen gas
e lead nitrate + hydrogen gas f hydrogen gas
g potassium acetate + hydrogen gas h sodium nitrate + water
i magnesium sulfate + water + carbon dioxide j calcium chloride + water + carbon dioxide
k iron acetate + water + carbon dioxide l hydrochloric acid + sodium
m sodium hydroxide + sulfuric acid
- 2 Reactants are on the left side of the arrow, products on the right.
a hydrochloric acid + calcium carbonate powder → carbon dioxide + water + calcium chloride
b sulfuric acid + potassium hydroxide → water + potassium sulfate
c iron + nitric acid → hydrogen gas + iron nitrate
d → copper chloride
e → hydrogen gas

14 Reactions that Fizz

- 2 a Carefully place a cork in the test tube with the collected gas.
Burn a wooden splint, then gently blow it out, making sure the splint is left glowing.
Take out the cork and place the glowing splint in the mouth of the test tube.
Observe what happens and record your results.
b Carefully light a match and hold it with your fingers.
Cork the test tube that contains the collected gas.
Hold the lit match above the test tube.
Record your observations.
c Collect the gas produced by bubbling it through some plastic tubing.
Place the other end of the plastic tubing into some limewater held in another test tube.
Make sure the tubing is not blocked.
Observe the limewater then record your results.

Name of gas	Method for test	Results
oxygen gas	Glow test: Hold a glowing splint into oxygen gas.	The glowing splint relights.
hydrogen gas	Pop test: Hold a burning match into the hydrogen gas.	Hydrogen gas explodes with a pop sound.
carbon dioxide	Limewater test: Bubble carbon dioxide through limewater.	Limewater goes milky white.

- 4 a calcium carbonate + hydrochloric acid → carbon dioxide gas + water + calcium chloride
b carbon dioxide + calcium hydroxide → water + calcium carbonate
c magnesium + hydrochloric acid → hydrogen gas + magnesium chloride
d hydrogen gas + oxygen gas → water vapour
e hydrogen peroxide → oxygen gas + water
f carbon + oxygen gas → carbon dioxide gas
- 5 a $\text{CaCO}_3 + \text{HCl} \rightarrow \text{CO}_2 + \text{H}_2\text{O} + \text{CaCl}_2$
b $\text{CO}_2 + \text{Ca(OH)}_2 \rightarrow \text{H}_2\text{O} + \text{CaCO}_3$
c $\text{Mg} + \text{HCl} \rightarrow \text{H}_2 + \text{MgCl}_2$
d $\text{H}_2 + \text{O}_2 \rightarrow \text{H}_2\text{O}$
e $\text{H}_2\text{O}_2 \rightarrow \text{O}_2 + \text{H}_2\text{O}$
f $\text{C} + \text{O}_2 \rightarrow \text{CO}_2$

15 To precipitate or not to precipitate

- 1 b silver chloride + potassium nitrate
c copper chloride + calcium nitrate



- d barium hydroxide + potassium nitrate
e lead sulfate + sodium nitrate
- 2 a white, magnesium carbonate
b no precipitate
c white, silver chloride
d green, copper carbonate
e white, barium sulfate

3

Positive ion	Negative ion			
	Carbonate (CO_3^{2-})	Chloride (Cl^-)	Hydroxide (OH^-)	Nitrate (NO_3^-)
calcium Ca^{2+}	CaCO_3	CaCl_2	Ca(OH)_2	$\text{Ca(NO}_3)_2$
copper Cu^{2+}	CuCO_3	CuCl_2	Cu(OH)_2	$\text{Cu(NO}_3)_2$
magnesium Mg^{2+}	MgCO_3	MgCl_2	Mg(OH)_2	$\text{Mg(NO}_3)_2$
silver Ag^+	Ag_2CO_3	AgCl	AgOH	AgNO_3
sodium Na^+	Na_2CO_3	NaCl	NaOH	NaNO_3

- 4 a 2 b 2 c 2 d 2 e 1 f 4
g 3 h 3 i eg: $\text{Ca(NO}_3)_2$, 9 j NaCl , 2

16 Burning it all

- 1 carbon + oxygen gas $\rightarrow$ carbon dioxide
- 2 a magnesium + oxygen gas $\rightarrow$ magnesium oxide, $\text{Mg} + \text{O}_2 \rightarrow \text{MgO}$
b $2 \text{Mg} + \text{O}_2 \rightarrow 2 \text{MgO}$
- 3 a hydrogen, carbon
b methane + oxygen gas $\rightarrow$ carbon dioxide + water
c $\text{CH}_4 + 2 \text{O}_2 \rightarrow \text{CO}_2 + 2 \text{H}_2\text{O}$
- 4 a methane + oxygen gas $\rightarrow$ carbon + water, $\text{CH}_4 + \text{O}_2 \rightarrow \text{C} + \text{H}_2\text{O}$
b $\text{CH}_4 + \text{O}_2 \rightarrow \text{C} + 2 \text{H}_2\text{O}$
c methane + oxygen gas $\rightarrow$ carbon dioxide + water
- 5 a octane + oxygen $\rightarrow$ carbon dioxide + water, $\text{C}_8\text{H}_{18} + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$
b $\text{C}_8\text{H}_{18} + 12\frac{1}{2} \text{O}_2 \rightarrow 8 \text{CO}_2 + 9 \text{H}_2\text{O}$ or better $2 \text{C}_8\text{H}_{18} + 25 \text{O}_2 \rightarrow 16 \text{CO}_2 + 18 \text{H}_2\text{O}$
c octane + oxygen gas $\rightarrow$ carbon dioxide + carbon monoxide + carbon + water, $\text{C}_8\text{H}_{18} + \text{O}_2 \rightarrow \text{CO}_2 + \text{CO} + \text{C} + \text{H}_2\text{O}$
d eg. $\text{C}_8\text{H}_{18} + 7 \text{O}_2 \rightarrow \text{CO}_2 + 3 \text{CO} + 4 \text{C} + 9 \text{H}_2\text{O}$ (These amounts vary depending on the amount of O_2 available.)
e Have the engine tuned, use correct petrol, run the engine when hot etc.
- 6 a carbon, hydrogen, oxygen
b glucose + oxygen gas $\rightarrow$ carbon dioxide + water, $\text{C}_6\text{H}_{12}\text{O}_6 + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$
c $\text{C}_6\text{H}_{12}\text{O}_6 + 6 \text{O}_2 \rightarrow 6 \text{CO}_2 + 6 \text{H}_2\text{O}$
d Glucose is part of food; it is digested in the stomach and small intestine, goes into the blood (blood plasma) and then to the body cells.
Oxygen is part of air; air is breathed in, the lungs absorb the oxygen and put it in the blood (in the red blood cells), the blood delivers it to the body cells.
e Carbon dioxide goes into the blood (blood plasma) from the cells, the blood carries it to the lungs, the lungs pass it out into the air as we breathe out.
Water from food and drink is passed into the blood, then either it is breathed out by the lungs, it is taken out by the kidneys and passed out with urine from the bladder, or it is perspired from special skin cells during sweating.
f Warm up and stretch before sport; cool down and stretch after sport; maintain a healthy level of fitness; keep drinking fluids during exercise.

17 Explaining acid rain

- 1 industrialised: countries with a lot of large-scale industry
is emitted: is given out
dissolves: mixes
a detrimental effect: a bad result
fish populations: number of fish
weathering: destruction of rocks by wind and water
- 2 a How acid rain gases are produced.
b How acid rain is formed.
c How acid rain affects things.
- 3 To explain the causes and effects of acid rain.
- 4 eg Para 1: they – the industrialised countries; para 3: it – acid rain
- 5 Various, eg the air, the former gas, the latter gas, the use of petrol; the atmosphere, the air, those acids, the rain drops; the acid rain, the long term effects, this rain, the pH, this, the soil, such effects.
- 6 iron + sulfuric acid $\rightarrow$ iron sulfate + hydrogen gas



- 7 calcium carbonate + sulfuric acid → calcium sulfate + water + carbon dioxide
 8 In Australia there is not as much burning of fossil fuels as in Europe and USA.
 9 Teacher to check.

18 An acid rain recount mix up

- 1 g - i - f - o - e (or e - o) - m - b - a - h - j - d - n - p - k - c - l
 2 Teacher to check.
 3 a factories b cars, trucks c burning fossil fuels
 d combusting petrol e sulfur dioxide f nitrogen dioxide
 g rain droplets h dissolution i sulfuric acid
 j nitric acid k acid rain l increase in acidity
 m destruction of motor n river, lakes o buildings
 p dead fish q collapsed buildings

20 pH and indicators

1

Substance	Colour of universal indicator
lemon	red
grapefruit	red or orange
cola drink	orange
strawberries	orange
milk	yellow
blood	green
eggs	green or blue
soap	blue
toothpaste	blue or purple
window cleaner	purple
oven cleaner	deep purple

2

Plant indicator	Litmus colour
hydrangea	blue
pansy	blue
pink geranium	blue
red cabbage	red
rhubarb	red
rose (pink)	blue
sweet pea	blue

- 3 a B b C, D, E c A

4

Solution	Red cabbage	Rhubarb	Rose
A	purple	red	yellow
B	red	red	pink
C	green	pale green	orange
D	green	pale green	orange
E	green	pale green	orange

- 5 pH is between 5 and 6.
 6 a B - F - A - D - C
 b The possible pH range is from 7 to 14, which is too broad. It is not possible to distinguish between a weak base and a strong base.
 7 a Cleaning substances are very basic and could damage the skin or mouth and stomach.
 b Rinse the eye with lots of water.
 8 Limestone must be basic since it neutralises the acidic soil. Adding basic limestone to the acidic soil makes the soil become neutral.

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